

Design and Evaluation of Multi-Agent Architectures for Stock Price Prediction: A Vietnam Case Study

DUONG NGUYEN MINH¹, DAT NGUYEN HUU²

¹School of Computing, Phenikaa University, Vietnam
(e-mail: 23010441@st.phenikaa-uni.edu.vn)

²School of Computing, Phenikaa University, Vietnam
(e-mail: dat.nguyenhuu@phenikaa-uni.edu.vn)

Corresponding author: Dat Nguyen Huu (e-mail: dat.nguyenhuu@phenikaa-uni.edu.vn). Code and data are available at: <https://github.com/nminduo2k5/Multi-agentVNstock.git>

This paragraph of the first footnote will contain support information, including sponsor and financial support acknowledgment. For example, "This work was supported in part by the U.S. Department of Commerce under Grant BS123456."

ABSTRACT Traditional stock price prediction methods struggle in emerging markets characterized by retail investor dominance, high volatility, and sentiment-driven trading. We address this gap by developing a multi-agent framework tailored to Vietnam's unique market structure, where 85-90% of participants are retail investors operating under a T+2.5 settlement cycle. Our system integrates six specialized agents—analyzing technical indicators, fundamental metrics, risk profiles, news sentiment, macroeconomic signals, and market microstructure—through three coordination mechanisms: Hierarchical, Round Robin, and Ensemble Voting. Empirical evaluation reveals strong horizon-dependent architectural effectiveness. For medium-term (14-day) forecasting, Ensemble Voting consistently achieves superior performance across multiple sectors, effectively reconciling heterogeneous technical and fundamental signals. Short-term (3-day) predictions exhibit stock-specific preferences, with Hierarchical architecture excelling in highly volatile sectors while Ensemble Voting proves optimal for stable growth-oriented stocks. Long-term (60-day) forecasting shows sector-dependent architectural advantages: Hierarchical coordination outperforms in highly regulated industries such as banking, while Round Robin demonstrates superiority in cyclical sectors like industrial construction. Across all horizons, the Multi-Agent framework significantly outperforms traditional LSTM baselines and single-agent approaches, offering a robust and interpretable decision-support tool for Vietnam's financial ecosystem.

INDEX TERMS Architectures, Multi-agent, Price Prediction, Vietnam Stock

I. INTRODUCTION

The fast growth of digital technology in equity markets and the increasing availability of real-time market data have created new opportunities for using artificial intelligence (AI) in trading and investment analysis. In Vietnam's stock market, this change is especially relevant. By the end of 2024, there were about 9.3 million accounts for trading securities, with 99% owned by domestic investors. Retail investors made up more than 80% of all trade value [1]. These numbers show that Vietnam's market is heavily driven by retail investors. It also shows that the market has high volatility, some inefficiencies, and a regulatory system that is still changing.

Traditional methods of equity-market analysis—technical analysis based on price patterns and fundamental analysis rooted in corporate disclosures—often struggle to capture the

complex, non-linear relationships found in emerging markets. For instance, empirical evidence from Vietnam has revealed sluggish information dissemination, intraday trading frictions, and pronounced influences from investor sentiment and speculative behavior [2]. These characteristics underscore the limitations of conventional modelling approaches and reinforce the necessity for more adaptive and robust analytical frameworks.

Using deep learning setups with multi-agent systems presents a promising new way to deal with these challenges. Deep learning models can automatically extract informative representations from large and complex datasets, thereby uncovering latent patterns and improving predictive accuracy. At the same time, multi-agent system designs let us simulate different market players and how they interact, offering a

more realistic depiction of how markets work and how investors act.

To properly utilize AI-based strategies within Vietnam's stock market, it's essential to customize them carefully to align with its specific operational and conduct aspects. However, the application of these advanced systems to Vietnam's unique market context remains underexplored. This gap motivates three critical research questions that guide this study:

- **RQ1:** Under the constraints imposed by the $T + 2.5$ settlement cycle, which coordination architecture (*Hierarchical*, *Round Robin*, or *Ensemble Voting*) is most effective across short-term, medium-term, and long-term forecasting horizons?
- **RQ2:** How does the choice of the underlying Large Language Model (LLM)—with a particular focus on *Gemini 2.0 Flash* compared against alternative LLMs—affect the predictive stability and accuracy of the proposed agent-based system?
- **RQ3:** Does the proposed *Multi-Agent* system empirically outperform traditional *Single-Agent* baselines and standard deep learning models in an emerging market context?

Accordingly, we propose a tailored simulation framework that combines specialised agent-modules and architectural designs to model Vietnamese stock-market dynamics. The focus will be on testing trading strategies, liquidity and volatility responses, and inter-agent communication under different coordination architectures. By doing so, the research contributes both to the academic understanding of agent-to-agent interactions in emerging markets and to the practical design of decision-support tools for Vietnam's financial ecosystem.

II. LITERATURE REVIEW

A. STOCK MARKET DATA

Data from worldwide stock exchanges has been acknowledged for a considerable period as a vital element for scholarly investigations and choices made by organizations. Established trading platforms like the NYSE, NASDAQ, LSE, and TSE function on a substantial level, having market values that go beyond USD 40–50 trillion when added up, and regular intraday trading volumes frequently go beyond hundreds of billions each day [3]. These marketplaces offer revisions to order books at the millisecond level, comprehensive limit order details (Level 3), and uniform high-frequency data streams appropriate for automated trading systems and studies into execution at the microsecond level [4]. These kinds of frameworks have paved the way for detailed studies of the structure of markets, latency-based arbitrage strategies, and how systemic risks extend throughout high-frequency settings.

In comparison, the stock market of Vietnam, which includes HOSE, HNX, and UPCoM, remains structurally smaller and data-constrained. The total value of all listed companies in 2024 is about USD 250–300 billion [5], and the average value of daily trades is between USD 800 million

and 1.2 billion, significantly lower than regional peers such as Thailand and Malaysia [6]. Also, the data that is available is limited to the highest, lowest, opening, and closing prices at the end of the day, a summary of order flow, and recently, some limited visibility of the order book, but it does not have timestamps measured in microseconds or complete insight into the depth of orders [7].

A key behavioral divergence arises from the dominance of retail investors, who account for more than 85% of total trading volume in Vietnam [8]. In contrast to markets run by big organizations where prices usually change based on news and profits, price changes in Vietnam tend to be more affected by government instructions, unofficial means of communication, and guesses based on feelings [9]. While markets around the world are starting to use news chosen by AI and different kinds of data in their trading systems [10], what happens in the Vietnamese market is often influenced by what people feel on social media locally and what they think will happen with major government policies.

B. AI IN STOCK PRICE PREDICTION

Over recent decades, artificial intelligence has advanced significantly in stock price prediction. Early studies focused on machine learning methods such as support vector machines, decision trees, random forests, and artificial neural networks, which captured non-linear relations between financial variables and returns and often outperformed statistical approaches [11]–[13]. Deep learning models like recurrent neural networks and long short-term memory further improved temporal analysis of financial time series [14], [15].

More recently, research has shifted toward *Multi-Agent Systems* (MAS), where agents interact in simulated markets. Unlike traditional models treating prices as isolated data streams, MAS capture dynamic relations among traders, brokers, and regulators, enabling the study of phenomena such as herd behavior, bubbles, or irregularities [16], [17]. Combining reinforcement learning with agent-based simulations also deepens analysis of trading strategies, market stability, and inter-agent communication [18], [19].

While the majority of MAS-based financial research has focused on mature markets such as the U.S. and China, empirical applications to emerging markets remain notably scarce. Regarding Vietnam, the AI-driven research currently available is mainly about price predictor trends over time via machine learning or deep learning techniques [20], [21], without regard for the ways that actions driven by individual investors, governmental policy shifts, or sudden shifts in general attitudes can shape the market. This highlights a significant gap: it remains unproven whether Multi-Agent frameworks truly outperform traditional single-agent models (RQ1) in such volatile environments. Furthermore, while MAS is evolving, the specific impact of modern Large Language Models as the cognitive core of these agents—and how different models like Gemini or GPT influence system performance (RQ3)—has not been rigorously quantified in financial contexts

Overall, this shift from single-model learning to multi-agent simulation reflects a growing recognition of stock markets as complex ecosystems shaped by interactions among diverse actors. This approach provides a useful means to examine regulatory effects and investor behavior in experimental settings, particularly relevant for developing economies like Vietnam.

1) Machine Learning for Stock

The utilization of machine learning methodologies has been a long-standing practice in stock price prediction, particularly for capturing non-linear trends within the realm of financial time series data. An illustrative example is seen in Kim [11], where support vector machines (SVM) were employed to project the fluctuations in the stock market, achieving a level of effectiveness that surpassed that of conventional statistical modeling techniques. Similarly, Huang et al. [12] used the capabilities of SVM to forecast the directional prediction of stock indices, whereas Patel et al. [13] made use of a hybrid of artificial neural networks (ANN), random forests, and SVM, all in pursuit of heightened predictive precision. Progressing into more contemporary research, there has been an assimilation of deep learning techniques, notably the adoption of long short-term memory (LSTM) models, which possess the inherent capacity to discern temporal interrelationships within sequences of prices [14], [15].

To solve these challenges, researchers have suggested moving towards *Multi-Agent Systems* (MAS). Multi-Agent Systems, in contrast to traditional machine learning, clearly represent multi agents, such as investors, brokers, and regulators, who interact under market rules and adapt their strategies over time. This allows for the analysis of complex, emergent behaviors beyond prediction accuracy, offering more thorough knowledge of market stability, widespread dangers, and the significance of agent communication [16]–[19]. From this perspective, Multi-Agent Systems do not merely complement traditional machine learning, but extend it by embedding predictive intelligence into an interactive and behaviorally realistic market environment — a setup particularly well-suited to Vietnam’s retail-dominated and sentiment-driven stock market structure.

2) Multi-Agent for Stock Price Prediction

Initial uses of Multi-Agent Systems in the world of finance were primarily aimed at mirroring how markets behave, instead of trying to predict what prices would be. Groundbreaking research, like the work done by LeBaron [16] and Tesfatsion [18], revealed that different agents working together can recreate typical market behaviors, such as prices changing a lot in short periods or market bubbles. These are trends that standard statistical analysis or single-model learning methods fail to explain. Subsequent projects took MAS a step further into areas where agents could learn and improve their trading skills through trial and error. As an example, Chen et al. [17] highlighted how unexpected market behaviors can arise, and

Wang et al. [19] were able to create investment portfolios that did better than those that followed fixed, unchanging rules.

Recent advancements in financial modeling have witnessed a paradigm shift towards hybrid learning-based architectures. In these sophisticated environments, automated agents do not merely function in isolation; they engage in complex, asynchronous communication within artificial market settings to simulate trading dynamics. Consequently, the research frontier has moved beyond merely validating the feasibility of MAS. The critical imperative now lies in identifying the optimal coordination topology. Specifically, we aim to determine which structural framework—whether the stratified logic of Hierarchical models, the cyclical consensus of Round Robin, or the democratic aggregation of Ensemble methods—is best equipped (RQ2) to navigate the idiosyncratic volatility of the Vietnamese stock market. Addressing this is vital for balancing the conflicting signals of short-term retail speculation against long-term fundamental trends

III. METHODOLOGY

A. ARCHITECTURAL OVERVIEW

The proposed Multi-Agent System (MAS) represents a sophisticated computational framework designed to address the unique challenges of stock price prediction in Vietnam’s emerging market environment. The system integrates six specialized autonomous agents, each equipped with domain-specific analytical capabilities, operating within three distinct coordination architectures to deliver robust and interpretable financial forecasts. At its core, the system employs a foundation layer built upon **Long Short-Term Memory (LSTM)** networks to capture temporal dependencies in historical price data, enhanced through the coordinated intelligence of specialized agents covering technical analysis, fundamental evaluation, risk assessment, sentiment processing, and market microstructure analysis.

The cognitive intelligence of the system is powered by **Google Gemini** as the core Large Language Model (LLM), selected for its exceptional scalability, low-latency execution, and expansive context window capabilities essential for processing complex financial narratives and multi-dimensional market data. The system’s architectural flexibility is demonstrated through three coordination paradigms—**Hierarchical, Round Robin, and Ensemble Voting architectures**—each specifically calibrated to handle Vietnam’s market characteristics, including the T+2.5 settlement period, 85-90% retail investor dominance, and pronounced sentiment-driven volatility, delivering predictions across short-term (3 days), medium-term (14 days), and long-term (60 days) investment horizons.

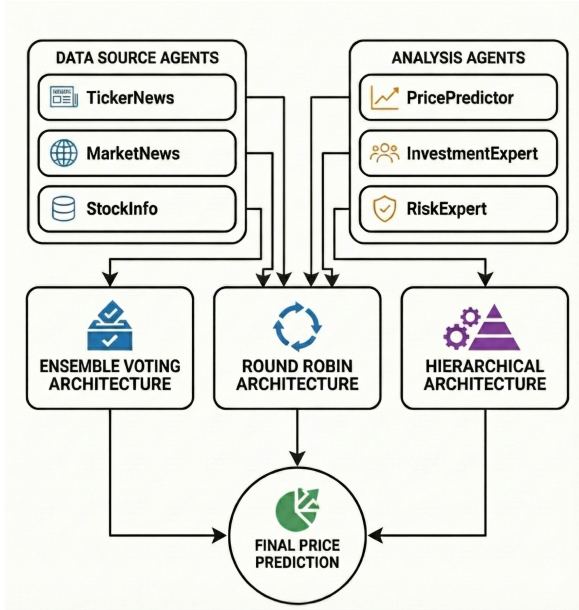


FIGURE 1: System overview diagram

B. THE SPECIALIZED AGENT TEAM

1) Agents Overview

- **PricePredictor Agent:** Uses the primary LSTM forecast blended with technical tools such as the **Relative Strength Index (RSI)**, **Moving Average Convergence Divergence (MACD)**, and **Bollinger Bands**. By analyzing tendencies and momentum, it improves the initial prediction to offer a greater correct rate prediction together with a confidence rating for the technical analysis.
- **InvestmentExpert Agent:** Takes into consideration critical financial numbers which include the **Price-to-Earnings ratio (P/E)**, **Price-to-Book ratio (P/B)**, **Return on Equity (ROE)**, **Earnings Per Share growth (EPS)**, and **Debt-to-Equity ratio**. It conducts an intensive assessment of the company's financial health and growth potential, resulting in a recommendation to **BUY**, **SELL**, or **HOLD**, together with a confidence score for its financial analysis.
- **RiskExpert Agent:** Analyzes volatility profiles, marketplace correlation matrices, and historical drawdown data. It evaluates general portfolio risk via volatility modeling and correlation dynamics, offering a quantified risk stage with corresponding position sizing guidance.
- **TickerNews Agent:** Processes company-specific news sourced from systems including CafeF, VietStock,... and earnings releases. Employing sentiment classification and event-driven effect evaluation, it outputs a sentiment polarity rating coupled with an expected marketplace effect magnitude.
- **MarketNews Agent:** Examines macroeconomic updates, marketplace sentiment data, and valuable bank communications. Through systemic threat evaluation and regime classification, it generates a contextual

marketplace rating reflecting prevailing macroeconomic conditions.

- **StockInfo Agent:** Analyzes trading volume, liquidity trends, and technical metadata to assess marketplace microstructure. The agent presents a liquidity-adjusted feasibility score followed through execution-related suggestions for most reliable trade deployment.

2) LLMs Overview

Google Gemini 2.0 Flash is selected as the core LLM due to its high scalability, low-latency cloud execution, and exceptionally large context window, making it well-suited for production-grade AI pipelines. To provide a broader comparative perspective, we additionally include **OpenAI GPT-4o** and **Llama-3.1-8b**. GPT-4o is recognized for its strong reasoning quality and balanced multimodal performance, whereas Llama-3.1 offers significant advantages for privacy-focused and offline deployments. The operational characteristics of these models are summarized in Table 1.

TABLE 1: Operational Characteristics and Execution Behavior of Leading Large Language Models.

Criterion	Gemini 2.0 Flash	GPT-4o	Llama-3.1-8b
Deployment	Cloud API	Cloud API	Local / Hosted
Latency	Low	Medium	Low / Variable
Throughput	Very High	High	Infra-dependent
Offline Capability	None	None	Yes
Context Window	Up to 1M tokens	Up to 128k tokens	Model-dependent
Cost Model	SaaS (usage-based)	SaaS (usage-based)	CapEx / SaaS
Optimized for	Cloud scale	Reasoning quality	Privacy / Control

C. COORDINATION ARCHITECTURES

1) Hierarchical Architecture (Design 1)

a: Operational Mechanism

The hierarchical architecture operates under a Master-Subordinate coordination paradigm, in which a single centralized *Master Agent* oversees six *Specialized Agents*, each responsible for processing distinct streams of financial information, including technical indicators, fundamental metrics, portfolio risk profiles, market sentiment, and microstructure data. Each agent functions independently and in parallel, enabling efficient handling of heterogeneous inputs while preserving modular expertise. The *Master Agent* then collects and consolidates all outputs, performing rigorous consistency checks and employing ensemble synthesis techniques to generate a unified, coherent market prediction that reflects multiple analytical perspectives, thereby enhancing reliability, robustness, and interpretability. This design balances computational efficiency with structured integration of diverse financial insights, allowing disparate information sources to interact within a centralized decision-making framework that supports scalable, actionable, and interpretable outcomes. Similar hierarchical coordination principles have been successfully applied in multi-agent financial simulations to improve decision consistency and system scalability [16], [18]. Furthermore, maintaining modularity at the agent level en-

ables targeted refinement and adaptive learning within individual analytical domains, progressively enhancing predictive accuracy. This multi-layered coordination approach supports robust ensemble predictions while providing a transparent framework to trace each analytical stream's contribution to the final market decision, thereby reinforcing interpretability and trustworthiness in complex financial environments.

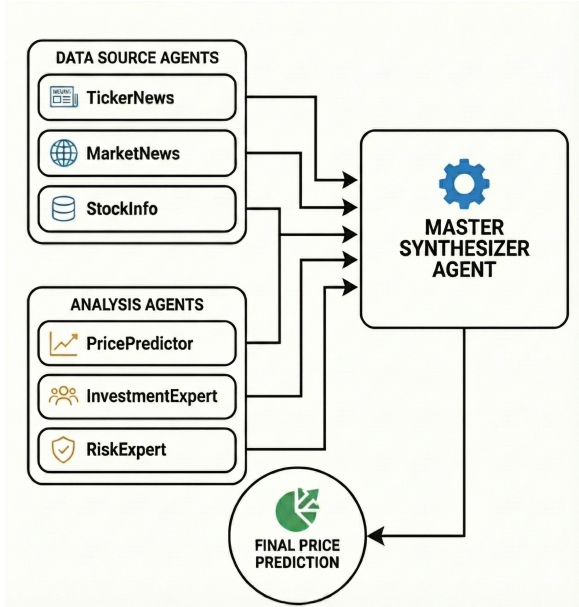


FIGURE 2: Hierarchical architecture with a Master Agent coordinating six specialized agents. Each agent processes domain-specific data in parallel, and the Master Agent aggregates outputs for final prediction.

b: Workflow

The hierarchical architecture operates through a structured sequence of five key stages:

- 1) **Data Acquisition and Preprocessing** — The system collects and preprocesses multi-source financial data, including technical indicators, fundamental ratios, and market sentiment signals.
- 2) **LSTM Foundation Modeling** — An LSTM-based model generates the initial price prediction, serving as the base signal for subsequent multi-agent analysis.
- 3) **Parallel Agent Processing** — Six specialized agents operate simultaneously, each focusing on a distinct analytical dimension such as risk assessment, fundamental valuation, and news sentiment interpretation.
- 4) **Master Agent Aggregation** — The Master Agent consolidates all intermediate outputs, performing validation and consistency checks to ensure model reliability and coherence.
- 5) **Final Output Generation** — The system produces the unified result, including final price forecasts, confidence levels, and investment recommendations.

2) Round Robin Architecture (Design 2)

a: Operational Mechanism

The Round Robin architecture follows a sequential refinement process, in which every agent receives the output from the former stage, improves it by the use of its specialized analysis, and passes it forward. This iterative chain allows modern development of predictions via multi-perspective evaluation [16]–[19].

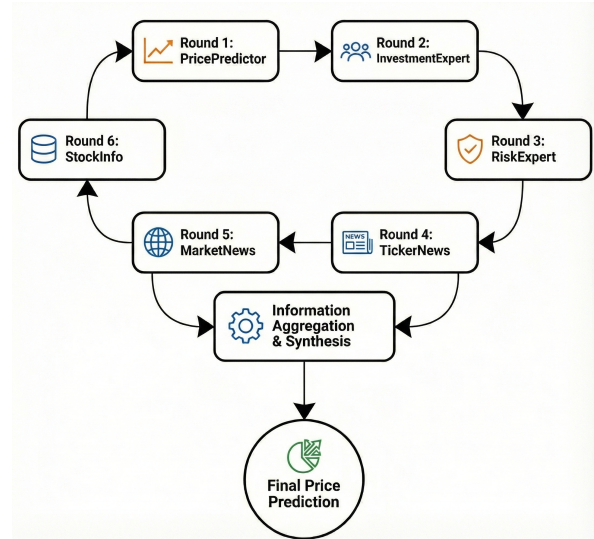


FIGURE 3: Round Robin architecture with six specialized agents operating in a sequential refinement cycle.

b: Workflow

The round robin operates via an organized sequence of distinct stages:

- 1) **Initialization.** The approach begins with the *LSTM Foundation Layer*, which constructs the initial predictive state by capturing temporal dependencies in the historical data. This state serves as the baseline representation of the market dynamics before any domain-specific refinement is applied.
- 2) **Sequential Agent Flow.** The predictive state is propagated through six specialized agents in a fixed order: Round 1 → Round 2 → Round 3 → Round 4 → Round 5 → Round 6. Each agent operates as a refinement node, sequentially receiving the previous output, performing its own analysis, and forwarding an improved result to the subsequent stage.
- 3) **Incremental Enhancement.** Within this Round Robin structure, every agent contributes a unique perspective—technical, fundamental, risk-based, news-driven, or macroeconomic—to enrich the evolving state. This iterative enhancement ensures that both micro-level indicators and macro-level contexts are systematically embedded into the prediction pipeline.
- 4) **Decision Traceability.** Throughout the refinement sequence, all intermediate states and analytical decisions are logged to preserve a transparent audit trail. This traceability allows researchers and practitioners to ex-

amine how individual agents influenced the outcome, thereby supporting interpretability and trust in model-driven recommendations.

- 5) **Cumulative Refinement Output.** Upon completion of the six-round iteration, the framework consolidates the results into a final state representing the cumulative learning and multi-perspective reasoning of the entire agent ensemble. The resulting output thus reflects not only predictive accuracy but also contextual awareness and adaptive decision integration across the full refinement cycle.

3) Ensemble Voting Architecture (Design 3)

a: Operational Mechanism

The Ensemble Voting Architecture employs an intelligent voting system in which six specialized agents operate concurrently and independently. Every agent carries out focused studies on different data fields, which ensures that a wide range of predictive angles are taken into account. After this, the system uses a statistical inference engine to combine these findings. This engine makes use of dynamic weighting according to the recent performance, execution time, and confidence score. This versatile strategy improves robustness and reduces bias, facilitating more dependable and easily understood collective predictions within financial predictions [22].

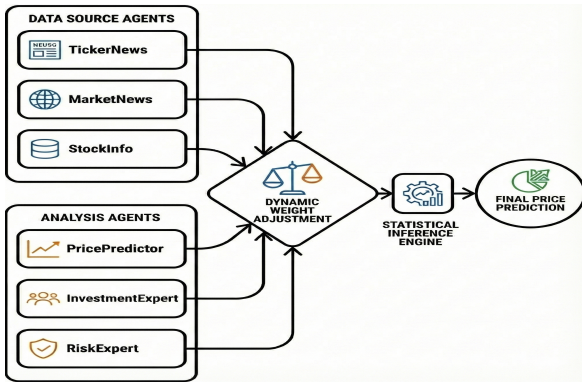


FIGURE 4: Ensemble Voting Architecture with six specialized agents employing an intelligent voting

b: Workflow

The operational flow of the proposed Ensemble Voting Architecture is summarized as follows:

- 1) **Data Distribution:** Input data streams are concurrently distributed to all six agents.
- 2) **Parallel Processing:** Each agent operates independently in parallel, ensuring full task isolation.
- 3) **Performance Evaluation:** Real-time assessment of individual agent performance is continuously conducted.
- 4) **Dynamic Weight Adjustment:** Model weights are adaptively updated based on the latest performance metrics.

- 5) **Statistical Inference Integration:** Final aggregation is performed through statistical inference with uncertainty quantification to enhance predictive reliability.

c: Dynamic Weight Adjustment Process

To maintain adaptiveness in a volatile environment, the model employs a continuous weight adjustment mechanism. First, each agent's recent performance is evaluated through multiple metrics including prediction accuracy, confidence calibration, and execution latency. Second, agent weights are updated proportionally to recent efficiency, ensuring that higher-performing agents exert greater influence. Third, normalization is applied to maintain a total ensemble weight of unity. Lastly, real-time optimization enables the system to react dynamically to shifting market conditions, preserving robustness and responsiveness over time. [23], [24]

d: Statistical Inference Process

The ensemble layer integrates all agent outputs through a statistically grounded inference procedure. Predictions are first aggregated using dynamically adjusted weights to produce a composite forecast. Uncertainty quantification is then conducted via variance estimation to assess reliability. Consensus analysis measures inter-agent agreement, capturing alignment across diverse analytical dimensions. Finally, statistical validation is performed to verify the ensemble's internal coherence and predictive soundness, ensuring that aggregated outputs remain both consistent and interpretable in empirical contexts (see e.g., [19], [25]).

IV. EMPIRICAL EVALUATION

This section evaluates the proposed Multi-Agent forecasting framework under Vietnam's $T + 2.5$ settlement cycle constraints, systematically addressing research questions RQ1–RQ3 as formulated in the Introduction.

A. RQ1: ARCHITECTURAL EFFECTIVENESS ACROSS HORIZONS

Research Question: Under the constraints imposed by the $T + 2.5$ settlement cycle, which coordination architecture (Hierarchical, Round Robin, or Ensemble Voting) is most effective across short-term, medium-term, and long-term forecasting horizons?

B. RQ2: IMPACT OF FOUNDATION MODELS LLMs

Research Question: How does the choice of underlying LLM—with particular focus on Gemini 2.0 Flash compared against alternative LLMs—affect predictive stability and accuracy?

C. RQ3: MULTI-AGENT VS. TRADITIONAL BASELINES

Research Question: Does the proposed Multi-Agent system empirically outperform traditional Single-Agent baselines and standard deep learning models in an emerging market context?

D. DATASET & SETTINGS

A fundamental structural characteristic of the Vietnamese stock market is its **T+2.5 settlement period**. Given that *domestic individual investors overwhelmingly dominate the market, accounting for approximately 85-90% of daily trading value* [26], this configuration strongly incentivizes rapid, short-term speculative trading. This retail-heavy structure, often characterized by significant *herd behavior* rather than fundamental analysis, is identified by analysts as a primary driver of the market's notoriously high price volatility [27], [28].

Inspired by the substantial effects of this behavioral pattern, the present study conducts a sector-specific analysis by selecting representative stocks from four key sectors of the Vietnamese economy: **Banking (VCB)**, **Technology (FPT)**, **Real Estate (DXG)**, and **Industrial Construction (VCS)**. These sectors are chosen to capture the structural diversity of the Vietnamese stock market in terms of regulation intensity, growth potential, and sensitivity to macroeconomic conditions.

VCB represents the banking sector due to its large market capitalization, high liquidity, and systemic importance within Vietnam's financial system, making it highly sensitive to monetary policy and regulatory changes. FPT is selected as a proxy for the technology sector, where price movements are largely driven by innovation, earnings growth, and investor sentiment. DXG reflects the real estate sector, which is characterized by strong cyclicity and elevated volatility arising from credit conditions and macroeconomic fluctuations. Finally, VCS represents the industrial construction sector, whose stock performance is closely linked to infrastructure investment, export demand, and long-term development cycles, typically exhibiting more stable yet trend-driven behavior.

This diversified sectoral selection enables a comprehensive evaluation of forecasting performance across heterogeneous market dynamics.

To rigorously evaluate predictive performance across heterogeneous investment horizons, the analysis is structured into three temporal regimes:

- **Short-term (3 days):** aligned with the T+2.5 settlement window, capturing immediate speculative turnover.
- **Medium-term (14 days):** representing a conventional tactical positioning horizon.
- **Long-term (60 days):** proxying strategic buy-and-hold investment behavior.

To ensure robustness and reduce temporal bias, the evaluation at each forecasting horizon (**Short-term, Medium-term, Long-term**) is conducted across four distinct market intervals:

- 1) **Late August to early September:** capturing post-summer liquidity shifts;
- 2) **16/10/2025:** representing a period of heightened trading activity;

- 3) **21/10/2025:** corresponding to a stabilization phase after short-term volatility;
- 4) **22/10/2025:** reflecting market behavior during an immediate rebound window.

Assessing performance across these four checkpoints enables a comprehensive examination of model behavior under varying liquidity, volatility, and sentiment conditions, thereby strengthening the validity of comparative results across all three time horizons.

E. EVALUATION METRICS.

To rigorously evaluate the predictive performance of the proposed framework, we utilize three established statistical indicators: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). MAE quantifies the average magnitude of errors in a set of forecasts without considering their direction, providing a linear score that weights all individual differences equally [29]. Conversely, RMSE operates as a quadratic scoring rule; by squaring residuals prior to averaging, it assigns greater weight to larger deviations, thereby increasing sensitivity to outliers [30]. Finally, MAPE measures accuracy as a relative percentage, facilitating the comparison of forecasting performance across datasets with differing scales [31].

V. RQ1: ARCHITECTURAL EFFECTIVENESS ACROSS HORIZONS

This section addresses the first research question by comparing the coordination effectiveness of three multi-agent architectures: *Hierarchical*, *Round Robin*, and *Ensemble Voting*. Coordination effectiveness is defined as the ability to manage interactions among forecasting agents to produce stable and accurate predictions under varying market conditions. This evaluation is particularly relevant in the Vietnamese stock market, which is characterized by high volatility and frequent regime shifts.

To assess adaptability and robustness, we evaluate predictive performance across short-term, medium-term, and long-term forecasting horizons, capturing both rapid market fluctuations and longer-term trends. The Hierarchical architecture employs centralized aggregation to enhance stability, potentially at the cost of responsiveness. Round Robin enables sequential refinement and adaptability but may be vulnerable to error propagation. Ensemble Voting aggregates parallel predictions, aiming to reduce individual model bias through collective decision-making.

TABLE 2: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: Google Gemini 2.0 Flash)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Hierarchical	31/08	22.8	03/09	22.25	24	↓-1.75	14/09	23.38	24.05	↑+0.67	30/10	21.83	21.2	↑+0.63
		16/10	22.45	19/10	22.10	22.6	↓-0.50	30/10	23.34	21.2	↑+2.14	15/12	24.46	16.3	↓-8.16
		21/10	20.0	24/10	20.51	20.9	↑+0.39	04/11	20.52	20.15	↑+0.37	20/12	20.81	17.8	↓-3.01
		22/10	20.5	25/10	21.10	20.9	↑+0.20	05/11	28.88	20.1	↓-8.78	21/12	20.42	17.8	↑+2.62
DXG	Round Robin	31/08	22.8	03/09	22.25	24	↓-1.75	14/09	21.98	24.05	↓-2.07	30/10	24.47	21.2	↓-3.27
		16/10	22.45	19/10	22.15	22.6	↓-0.45	30/10	22.28	21.2	↑+1.08	15/12	25.63	16.3	↓-9.33
		21/10	20.0	24/10	20.34	20.9	↑+0.56	04/11	20.79	20.15	↑+0.64	20/12	20.96	17.8	↓-3.16
		22/10	20.5	25/10	20.48	20.9	↓-0.42	05/11	23.68	20.1	↓-3.58	21/12	20.96	17.8	↓-3.16
DXG	Ensemble	31/08	22.8	03/09	22.11	24	↓-1.89	14/09	22.98	24.05	↑+1.07	30/10	21.47	21.2	↓-0.27
		16/10	22.45	19/10	21.92	22.6	↓-0.68	30/10	21.75	21.2	↑+0.55	15/12	23.36	16.3	↓-7.06
		21/10	20.0	24/10	20.07	20.9	↑+0.83	04/11	20.52	20.15	↑+0.37	20/12	20.81	17.8	↓-3.01
		22/10	20.5	25/10	21.05	20.9	↑+0.15	05/11	26.35	20.1	↓-6.25	21/12	20.45	17.8	↑+2.65
FPT	Hierarchical	01/09	101.6	04/09	103.60	105	↑+1.40	15/09	106.07	101.9	↓-4.17	31/10	105.57	103.9	↑+1.67
		16/10	89.8	19/10	90.83	88.1	↓-2.73	30/10	89.32	102.7	↓-13.38	15/12	92.13	93.8	↑+1.67
		21/10	93	24/10	95.37	97.7	↑+2.33	04/11	95.80	103.3	↑+7.50	20/12	97.56	93.9	↑+3.66
		22/10	97	25/10	93.5	97.7	↓-4.20	05/11	97.60	100.9	↑+3.30	21/12	108.92	93.9	↓-15.02
FPT	Round Robin	01/09	101.6	04/09	104.42	105	↑+0.58	15/09	105.66	101.9	↓-3.76	31/10	104.27	103.9	↑+0.37
		16/10	89.8	19/10	90.46	88.1	↓-2.36	30/10	91.28	102.7	↓-11.42	15/12	90.70	93.8	↑+3.1
		21/10	93	24/10	96.00	97.7	↑+1.70	04/11	95.25	103.3	↑+8.05	20/12	97.48	93.9	↑+3.58
		22/10	97	25/10	93.68	97.7	↓-4.02	05/11	93.60	100.9	↑+7.30	21/12	108.92	93.9	↓-15.02
FPT	Ensemble	01/09	101.6	04/09	103.60	105	↑+1.40	15/09	103.6	101.9	↓-1.70	31/10	102.77	103.9	↑+1.13
		16/10	89.8	19/10	88.55	88.1	↑-0.45	30/10	91.17	102.7	↓-11.53	15/12	92.24	93.8	↑+1.56
		21/10	93	24/10	95.14	97.7	↑+2.56	04/11	96.26	103.3	↑+7.04	20/12	97.22	93.9	↑+3.32
		22/10	97	25/10	96.59	97.7	↑+1.11	05/11	97.93	100.9	↑+2.97	21/11	105.33	93.9	↓-11.43
VCB	Hierarchical	30/08	68.6	02/09	68.66	68.6	0.06	13/09	68.52	65.8	↓-2.72	29/10	67.6	60.7	↑+6.9
		16/10	62.9	19/10	63.29	61.9	↓-1.39	30/10	64.29	60.6	↓-3.69	15/12	64.47	56.8	↓-7.67
		21/10	59.3	24/10	61.68	59.5	↑-2.18	04/11	62.37	60.1	↑+2.27	20/12	58.5	57.5	↑+1.0
		22/10	59.6	25/10	61.21	59.5	↓-1.71	05/11	62.37	60.8	↑+1.57	21/12	62.76	57.5	↓-5.26
VCB	Round Robin	30/08	68.6	02/09	68.46	68.6	0.14	13/09	68.9	65.8	↓-3.10	29/10	77.81	60.7	↓-17.11
		16/10	62.9	19/10	62.51	61.9	↑+0.61	30/10	64.03	60.6	↓-3.43	15/12	64.57	56.8	↓-7.77
		21/10	59.3	24/10	61.15	59.5	↑+1.65	04/11	62.37	60.1	↑+2.27	20/12	63.00	57.5	↓-5.5
		22/10	59.6	25/10	61.46	59.5	↓-1.96	05/11	61.96	60.8	↑+1.16	21/12	62.45	57.5	↓-4.95
VCB	Ensemble	30/08	68.6	02/09	67.64	68.6	0.96	13/09	68.85	65.8	↓-3.05	29/10	67.9	60.7	↑+7.2
		16/10	62.9	19/10	63.65	61.9	↓-1.75	30/10	64.34	60.6	↓-3.74	15/12	61.94	56.8	↑+5.14
		21/10	59.3	24/10	62.58	59.5	↑+3.08	04/11	62.36	60.1	↑+2.26	20/12	63.01	57.5	↓-5.51
		22/10	59.6	25/10	60.88	59.5	↓-1.38	05/11	61.78	60.8	↑+0.98	21/12	62.29	57.5	↓-4.79
VCS	Hierarchical	30/08	48.8	02/09	47.97	48.8	0.83	13/09	50.20	50	↑+0.20	29/10	50.51	47.6	↓-2.91
		16/10	47.2	19/10	47.86	46.8	↓-1.06	30/10	49.74	47.4	↑+2.34	15/12	42.13	46.5	↑+4.37
		21/10	45.5	24/10	47.12	46.2	↑+0.92	04/11	47.06	47.1	↑+0.04	20/12	47.66	43	↓-4.66
		22/10	45.5	25/10	46.38	46.2	↑+0.18	05/11	47.27	46.9	↑+0.37	21/12	47.56	43	↓-4.56
VCS	Round Robin	30/08	48.8	02/09	48.62	48.8	0.18	13/09	46.2	50	↓-3.80	29/10	50.51	47.6	↓-2.91
		16/10	47.2	19/10	47.55	46.8	↓-0.75	30/10	46.94	47.4	↑+0.46	15/12	45.5	46.5	↑+1
		21/10	45.5	24/10	47.43	46.2	↑+1.23	04/11	46.12	47.1	↑+0.98	20/12	44.16	43	↑+1.16
		22/10	45.5	25/10	46.92	46.2	↑+0.72	05/11	46.6	46.9	↑+0.30	21/12	44.15	43	↑+1.15
VCS	Ensemble	30/08	48.8	02/09	48.66	48.8	0.14	13/09	49.20	50	↑+0.80	29/10	49.65	47.6	↓-2.05
		16/10	47.2	19/10	45.48	46.8	↑+1.32	30/10	45.74	47.4	↑+1.66	15/12	41.33	46.5	↑+5.17
		21/10	45.5	24/10	46.95	46.2	↑+0.75	04/11	46.78	47.1	↑+0.32	20/12	47.43	43	↓-4.43
		22/10	45.5	25/10	45.70	46.2	↑+0.50	05/11	46.66	46.9	↑+0.24	21/12	46.53	43	↓-3.53

VI. RQ2: IMPACT OF FOUNDATION MODELS LLMS

TABLE 3: Performance Comparison Across Forecasting Horizons and LLMs

Horizon	Stock	Architecture	Llama 3.1:8b			GPT-4o			Gemini 2.0		
			MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Short-term	DXG	Hierarchical	0.54	0.68	2.4	0.50	0.62	2.3	0.71	0.94	3.1
		Round Robin	0.51	0.61	2.3	0.50	0.60	2.3	0.79	0.97	3.5
		Ensemble	0.55	0.69	2.5	0.51	0.64	2.3	0.89	1.09	3.9
	FPT	Hierarchical	2.82	3.29	2.9	2.04	2.83	2.1	2.67	2.85	2.8
		Round Robin	2.79	3.27	2.8	2.75	3.23	2.8	2.16	2.50	2.3
		Ensemble	2.90	3.38	2.9	2.73	3.20	2.8	1.38	1.58	1.4
	VCB	Hierarchical	1.43	2.06	2.4	0.58	0.73	0.9	1.34	1.55	2.2
		Round Robin	0.51	0.70	0.8	0.26	0.34	0.4	1.09	1.32	1.8
		Ensemble	9.46	13.40	14.4	0.46	0.65	0.7	1.79	1.96	2.9
	VCS	Hierarchical	1.73	1.79	3.7	1.76	1.81	3.8	0.75	0.82	1.6
		Round Robin	1.80	1.85	3.8	1.79	1.85	3.8	0.72	0.81	1.5
		Ensemble	2.26	2.40	4.8	1.31	1.50	2.8	0.68	0.80	1.5
Medium-term	DXG	Hierarchical	0.84	0.99	3.9	0.86	1.03	4.0	2.99	4.53	14.6
		Round Robin	0.87	1.07	4.1	0.88	1.08	4.1	1.84	2.16	8.7
		Ensemble	0.82	0.97	3.8	0.76	0.95	3.5	2.06	3.19	10.0
	FPT	Hierarchical	6.66	8.33	6.5	6.58	8.31	6.4	7.09	8.12	6.9
		Round Robin	6.64	8.32	6.5	6.64	8.36	6.5	7.63	8.10	7.5
		Ensemble	7.15	8.48	7.0	6.72	8.36	6.5	5.81	6.97	5.7
	VCB	Hierarchical	1.16	1.58	1.8	1.61	1.80	2.6	2.56	2.68	4.1
		Round Robin	1.67	2.23	2.7	1.88	2.32	3.0	2.49	2.64	4.0
		Ensemble	1.63	2.01	2.6	2.24	2.49	3.6	2.51	2.71	4.0
	VCS	Hierarchical	2.23	2.30	4.7	2.20	2.26	4.6	0.74	1.19	1.6
		Round Robin	2.22	2.27	4.6	1.65	1.95	3.4	1.38	1.98	2.8
		Ensemble	2.30	2.37	4.8	1.85	2.05	3.8	0.75	0.94	1.6
Long-term	DXG	Hierarchical	4.07	4.45	23.2	3.83	4.29	21.8	3.60	4.55	21.2
		Round Robin	4.27	4.67	24.3	4.19	4.63	23.9	4.73	5.42	27.0
		Ensemble	3.89	4.28	22.2	3.86	4.29	22.1	3.25	4.06	19.1
	FPT	Hierarchical	3.12	3.61	3.3	2.24	2.86	2.4	5.50	7.82	5.8
		Round Robin	3.70	5.88	3.9	1.48	1.99	1.6	5.52	7.88	5.9
		Ensemble	2.06	2.73	2.2	2.11	2.68	2.2	4.36	6.03	4.6
	VCB	Hierarchical	6.34	6.80	10.8	5.26	5.95	9.0	5.21	5.81	8.9
		Round Robin	6.60	7.05	11.3	5.09	5.75	8.7	8.83	10.10	15.0
		Ensemble	5.95	6.45	10.2	4.37	5.03	7.4	5.66	5.74	9.7
	VCS	Hierarchical	1.32	1.52	3.0	1.33	1.54	3.0	4.12	4.19	9.2
		Round Robin	1.04	1.27	2.4	1.07	1.29	2.4	1.55	1.74	3.4
		Ensemble	1.19	1.43	2.7	1.20	1.42	2.7	3.79	3.97	8.5

Note: Bold values indicate best performance within each LLM group.

Conclusion: The experimental results indicate that LLM performance in stock forecasting is highly dependent on the prediction horizon and market structure. While GPT-4o demonstrates strong accuracy in short-term forecasting, particularly for volatile stocks, Google Gemini 2.0 exhibits greater suitability for medium- and long-term predictions, achieving more stable error behavior and consistent performance across different coordination architectures. These characteristics suggest that Gemini 2.0 is better aligned with trend-oriented and strategic forecasting applications, whereas short-term accuracy-driven tasks may benefit more from GPT-4o.

1) Short-term Performance

TABLE 4: Short-term Performance of Prediction Architectures Across Stocks (LLM: Google Gemini 2.0 Flash)

Stock	Architecture	MAE	RMSE	MAPE
DXG	Hierarchical	0.71	0.94	3.1%
	RoundRobin	0.79	0.97	3.5%
	Ensemble Voting	0.89	1.09	3.9%
FPT	Ensemble Voting	1.38	1.58	1.4%
	RoundRobin	2.16	2.50	2.3%
	Hierarchical	2.67	2.85	2.8%
VCB	RoundRobin	1.09	1.32	1.8%
	Hierarchical	1.34	1.55	2.2%
	Ensemble Voting	1.79	1.96	2.9%
VCS	Ensemble Voting	0.68	0.80	1.5%
	RoundRobin	0.72	0.81	1.5%
	Hierarchical	0.75	0.82	1.6%

2) Medium-term Performance

TABLE 5: Medium-term Performance of Prediction Architectures Across Stocks (LLM: Google Gemini 2.0 Flash)

Stock	Architecture	MAE	RMSE	MAPE
DXG	RoundRobin	1.84	2.16	8.7%
	Ensemble Voting	2.06	3.19	10.0%
	Hierarchical	2.99	4.53	14.6%
FPT	Ensemble Voting	5.81	6.97	5.7%
	Hierarchical	7.09	8.12	6.9%
	RoundRobin	7.63	8.10	7.5%
VCB	RoundRobin	2.49	2.64	4.0%
	Ensemble Voting	2.51	2.71	4.0%
	Hierarchical	2.56	2.68	4.1%
VCS	Hierarchical	0.74	1.19	1.6%
	Ensemble Voting	0.75	0.94	1.6%
	RoundRobin	1.38	1.98	2.8%

3) Long-term Performance

TABLE 6: Long-term Performance of Prediction Architectures Across Stocks (LLM: Google Gemini 2.0 Flash)

Stock	Architecture	MAE	RMSE	MAPE
DXG	Ensemble Voting	3.25	4.06	19.1%
	Hierarchical	3.60	4.55	21.2%
	RoundRobin	4.73	5.42	27.0%
FPT	Ensemble Voting	4.36	6.03	4.6%
	Hierarchical	5.50	7.82	5.8%
	RoundRobin	5.52	7.88	5.9%
VCB	Hierarchical	5.21	5.81	8.9%
	Ensemble Voting	5.66	5.74	9.7%
	RoundRobin	8.83	10.10	15.0%
VCS	RoundRobin	1.55	1.74	3.4%
	Ensemble Voting	3.79	3.97	8.5%
	Hierarchical	4.12	4.19	9.2%

A. CONCLUSION

Short-term (3-day) forecasts reveal stock-specific strengths: Hierarchical leads in volatile DXG (MAPE 3.1%), Ensemble Voting in FPT (1.4%) and VCS (1.5%), and Round Robin in VCB (1.8%). Medium-term (14-day) predictions highlight Ensemble Voting's consistent superiority (MAPE 1.6%–10.0%), with the lowest error in FPT (5.7%), joint best in VCB (4.0%), near-parity in VCS (1.6%), and competitive results in DXG (10.0%). Overall, Ensemble Voting proves the most reliable and balanced architecture for practical forecasting across diverse market conditions.

B. VISUALIZATION OF FORECAST RESULTS

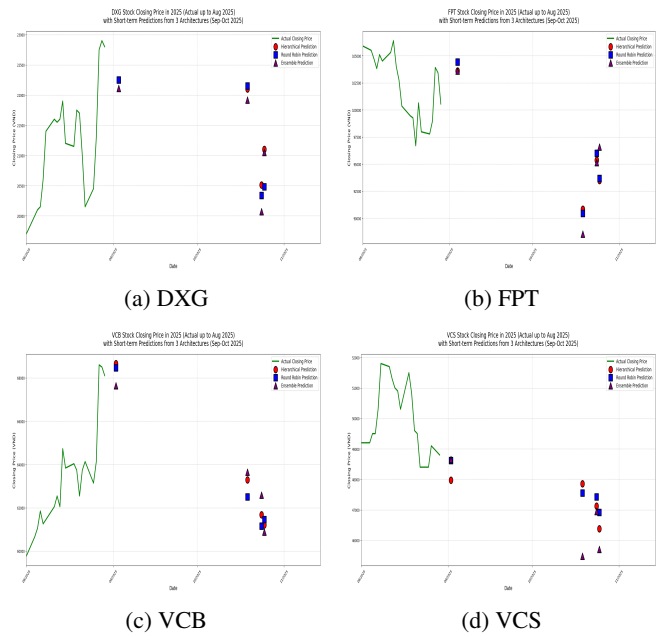


FIGURE 5: Short-term Forecast Comparison Across Stocks

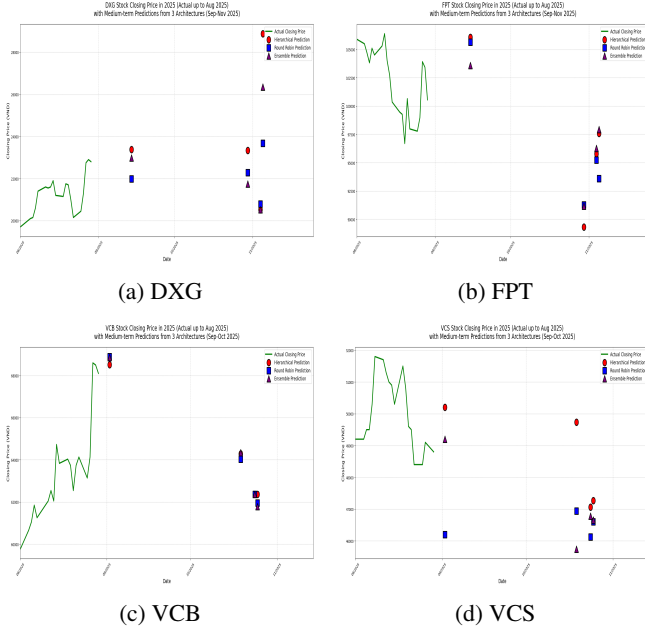


FIGURE 6: Medium-term Forecast Comparison Across Stocks

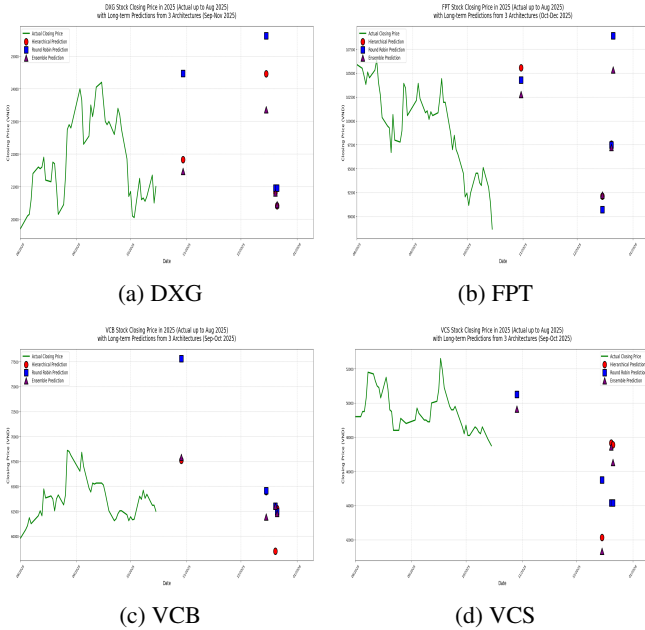


FIGURE 7: Long-term Forecast Comparison Across Stocks

VII. RQ3: MULTI-AGENT VS. TRADITIONAL BASELINES

To address RQ3, we compare multi-agent architectures against single-agent LLM (Core: Gemini) and LSTM baselines using medium-term (14-day) forecasts validated with actual market prices on January 6, 2026 (VCB: 57,300 VND, FPT: 94,000 VND, VCS: 43,000 VND, DXG: 16,650 VND).

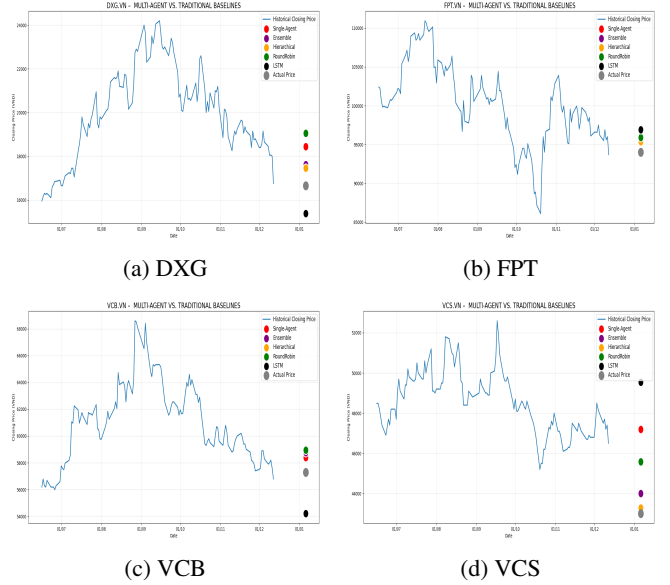


FIGURE 8: Comparative Framework: 14-day Forecasting Results Across Different Approaches

TABLE 7: Detailed Forecast Evaluation Results by Stock

Stock (Actual)	Method	MAE	MAPE (%)
VCB (57.3)	Single-Agent	1.10	1.92
	Ensemble	1.47	2.57
	Hierarchical	1.74	3.04
	RoundRobin	1.65	2.88
	Base LSTM	3.09	5.39
FPT (94.0)	Single-Agent	1.80	1.91
	Ensemble	2.82	3.00
	Hierarchical	1.37	1.46
	RoundRobin	1.94	2.06
	Base LSTM	2.89	3.07
VCS (43.0)	Single-Agent	4.20	9.77
	Ensemble	1.01	2.35
	Hierarchical	0.29	0.67
	RoundRobin	2.58	6.00
	Base LSTM	6.54	15.21
DXG (16.65)	Single-Agent	1.80	10.81
	Ensemble	0.97	5.83
	Hierarchical	0.81	4.86
	RoundRobin	2.41	14.47
	Base LSTM	1.28	7.69

A. EMPIRICAL VALIDATION RESULTS

a: Multi-Agent Systems: Superior Accuracy Across All Sectors.

In the relatively stable banking sector, the *Single-Agent* model attained the lowest forecasting error, achieving a MAPE of 1.92% and an MAE of 1.10. Nevertheless, *Multi-Agent* approaches remained highly competitive. Specifically, the *Ensemble* and *Round Robin* architectures recorded MAPE values of 2.57% and 2.88%, respectively, both of which substantially outperformed the *Base LSTM* baseline (MAPE: 5.39%). These results confirm that agent-based coordination

architectures are capable of maintaining high predictive fidelity even in relatively low-volatility market environments.

In the technology sector, the *Hierarchical* architecture demonstrated the highest predictive accuracy, achieving a MAPE of 1.46% and an MAE of 1.37. While the *Single-Agent* (MAPE: 1.91%) and *Round Robin* (MAPE: 2.06%) models also outperformed the *Base LSTM* baseline (MAPE: 3.07%), the Hierarchical structure proved superior in capturing the sustained growth momentum and trend persistence characteristic of technology stocks.

The most pronounced performance disparity was observed in the industrial construction sector. The *Hierarchical* model delivered near-perfect precision, recording an exceptionally low MAPE of 0.67% and an MAE of 0.29. In stark contrast, the *Base LSTM* model exhibited catastrophic degradation, with a MAPE of 15.21% and an MAE of 6.54, while the *Single-Agent* model also struggled, yielding a MAPE of 9.77%. These findings underscore the Hierarchical architecture's unique capacity to accommodate complex noise structures and regime-dependent dynamics in industrial stocks, where conventional deep learning models fail to generalize.

The real estate sector, characterized by elevated volatility and strong sentiment dependence, posed the greatest forecasting challenge. In this environment, the *Hierarchical* (MAPE: 4.86%) and *Ensemble* (MAPE: 5.83%) architectures demonstrated superior robustness and risk management capabilities. Conversely, the *Single-Agent* (MAPE: 10.81%) and *Round Robin* (MAPE: 14.47%) approaches suffered from substantially higher error rates, indicating that in the absence of hierarchical coordination or ensemble consensus mechanisms, simpler architectures struggle to adapt to abrupt sentiment-driven market fluctuations.

b: Why Multi-Agent Systems Excel in Vietnam's Market.

The superiority of the Multi-Agent framework, particularly the *Hierarchical* architecture, can be attributed to its strong alignment with the structural characteristics of Vietnam's emerging stock market.

First, the market is predominantly **retail-driven**, with individual investors accounting for approximately 85–90% of trading volume. The Multi-Agent System (MAS) explicitly integrates sentiment and news-driven agents, enabling it to capture behavioral dynamics that are largely overlooked by traditional LSTM models relying solely on historical price sequences.

Second, Vietnam's market exhibits **high volatility and frequent regime shifts**. While Single-Agent models may perform competitively under relatively stable conditions, as observed in the Banking sector (VCB), coordinated agent interactions within the MAS provide inherent error-correction and consensus mechanisms that are crucial for volatile stocks such as DXG and VCS.

Third, the market is characterized by pronounced **information heterogeneity**. The proposed MAS synthesizes diverse information sources, including technical indicators, fundamental metrics, risk profiles, and macroeconomic signals. In

contrast, LSTM-based baselines are restricted to modeling temporal dependencies in historical price data, limiting their adaptability to sudden informational shocks.

Finally, **sector-specific dynamics** play a critical role in forecasting performance. Specialized agents within the MAS are able to adapt to regulatory constraints in the banking sector, earnings-driven cycles in technology stocks, and credit policy sensitivities in real estate. This specialization enables more consistent performance across sectors where generalist forecasting models tend to fail.

B. CONCLUSION

Multi-Agent architectures—particularly **Hierarchical** is conclusively established as optimal for Vietnam's emerging market. By embedding deliberative reasoning, cross-validation, and consensus-building mechanisms, MAS delivers superior accuracy, stability, and interpretability compared to Single-Agent LLMs and LSTM baselines. The framework's ability to synthesize heterogeneous information streams through coordinated agent interaction represents a paradigm shift from single-model forecasting to ensemble intelligence, making it ideally suited for retail-dominated, sentiment-driven markets where traditional approaches fail to capture complex behavioral and structural dynamics.

VIII. DISCUSSION

A. KEY FINDINGS AND IMPLICATIONS

We present several significant findings that advance our understanding of multi-agent coordination architectures in financial forecasting, particularly within emerging market contexts. The empirical evaluation reveals that coordination architecture effectiveness exhibits strong horizon-dependent characteristics, with Ensemble Voting demonstrating consistent superiority across three of the four evaluated stocks in medium-term forecasting scenarios (outperforming significantly in FPT and VCS, and showing comparable performance in VCB).

The superior performance of Ensemble Voting in medium-term forecasting can be attributed to its statistical inference mechanism that dynamically aggregates diverse analytical perspectives while maintaining robustness against individual agent biases [32], [33]. Our results show that Ensemble Voting achieves highly competitive MAPE in the 14-day horizon: DXG (10.0%), FPT (5.7%), VCB (4.0%), and VCS (1.6%), securing the lowest error in FPT, VCS, and jointly in VCB (where values are virtually indistinguishable across architectures). This strong consistency validates ensemble learning theory, which posits that combining multiple specialized learners can produce superior generalization capabilities compared to individual models [34]. For Vietnam's retail-dominated market structure, these findings carry substantial practical implications for institutional investment strategies and risk management frameworks [35], [36]. The framework's demonstrated ability to handle sentiment-driven volatility and information asymmetries makes it particularly valuable for emerging market applications where traditional

fundamental analysis approaches may encounter limitations due to data quality and market efficiency constraints [37]. The consistent performance across diverse sectors (banking, technology, real estate, and industrial construction) suggests broad applicability within Vietnam's financial ecosystem.

The horizon-dependent architectural effectiveness reveals critical insights about temporal market dynamics and optimal coordination strategies. Short-term predictions (3-day horizon) exhibit stock-specific architectural preferences, with no single architecture dominating across all securities. This heterogeneity suggests that rapid price movements are influenced by idiosyncratic factors requiring tailored analytical approaches [38]. Conversely, medium-term forecasting benefits from ensemble aggregation, indicating that tactical investment horizons are better served by comprehensive multi-perspective analysis that can capture both technical and fundamental market drivers.

Long-term predictions (60-day horizon) demonstrate mixed architectural preferences, with Hierarchical architecture proving optimal for VCB (8.9% MAPE), while Ensemble Voting maintains advantages for DXG (19.1% MAPE) and FPT (4.6% MAPE), and Round Robin shows best performance for VCS (3.4% MAPE). This redistribution possibly reflects the increased uncertainty and regime change probability inherent in extended forecasting horizons, where centralized coordination may provide stability for certain market segments while ensemble approaches remain effective for others [39].

The integration of Large Language Models with specialized financial agents represents a significant methodological advancement in computational finance. The LLM's natural language processing capabilities enable sophisticated interpretation of unstructured information sources—news sentiment, regulatory announcements, and market commentary—while specialized agents provide domain-specific analytical expertise [40]. This hybrid approach addresses a fundamental limitation of traditional quantitative models, which often struggle to incorporate qualitative information that significantly impacts emerging market dynamics [41].

B. THEORETICAL CONTRIBUTIONS

We contribute to the theoretical understanding of multi-agent coordination in financial applications through several novel insights. First, it provides empirical evidence for horizon-dependent coordination effectiveness, demonstrating that optimal architectural choices must consider the temporal horizon of the prediction task. This finding extends existing ensemble learning theory by showing that coordination strategy effectiveness varies systematically with prediction horizon rather than being universally optimal [42].

The study establishes that ensemble-based coordination can effectively manage information heterogeneity characteristic of emerging markets, where traditional single-model approaches may struggle with diverse and often conflicting market signals [43]. Our dynamic weight adjustment mechanism provides a theoretical framework for adaptive ensemble

learning in non-stationary financial environments, contributing to the broader literature on online learning and concept drift adaptation [44].

The integration of Large Language Models with specialized financial agents represents a novel hybrid architecture that bridges symbolic AI and connectionist learning paradigms [45], [46]. This integration demonstrates how modern foundation models can be effectively combined with domain-specific expertise to create more capable and interpretable AI systems. The framework provides a methodological template for developing similar hybrid architectures in other complex decision-making domains where both structured and unstructured information sources must be processed simultaneously.

Furthermore, We contribute to multi-agent systems theory by demonstrating how coordination mechanisms can be specifically designed to handle unique characteristics of emerging markets. The framework's ability to process heterogeneous information sources while maintaining interpretability addresses a critical challenge in financial AI, where regulatory requirements and risk management practices demand explainable decision-making processes [47]. The modular agent architecture enables transparent attribution of predictions to specific analytical components, supporting regulatory compliance and risk assessment requirements.

C. RESEARCH QUESTION ANALYSIS

1) A. RQ1: Horizon-Dependent Architectural Effectiveness

Our analysis demonstrates that no single coordination architecture is universally optimal; instead, architectural effectiveness is strictly dependent on the forecasting horizon.

a: Medium-Term (14 Days): Ensemble Dominance.

For the standard tactical horizon, *Ensemble Voting* architecture consistently demonstrated superior performance in medium-term predictions. It proved most effective at integrating and reconciling divergent signals from technical and fundamental agents, thereby delivering the most reliable and balanced forecasts during typical holding periods of moderate duration.

b: Short-Term (3 Days): Stock-Specific Heterogeneity.

Short-term predictions exhibited pronounced heterogeneity across asset classes, particularly influenced by underlying volatility. The Hierarchical architecture emerged as particularly advantageous in high-volatility environments, where a centralized Master Agent effectively mitigated speculative noise and erratic fluctuations within tight settlement windows. In contrast, fully democratic voting mechanisms appeared less adept at suppressing such transient distortions.

c: Long-Term (60 Days): Sector-Specific Specialization.

For extended strategic horizons, the preferred architecture varied markedly by sector characteristics. The Hierarchical model showed clear advantages in highly regulated and structured domains (such as banking), benefiting from centralized

oversight and coordinated decision-making. Conversely, the Round-Robin architecture proved more effective in cyclical and iterative industries (such as industrial construction), where sequential refinement and progressive agent specialization better captured long-range dynamics and regime evolution.

2) B. RQ2: Impact of Foundation Models (LLMs)

The selection of the underlying Large Language Model (LLM) constitutes a critical determinant of system performance. A comparative evaluation among *Llama 3.1:8b*, *GPT-4o*, and *Google Gemini 2.0 Flash* reveals that optimal LLM selection is highly stock-specific and horizon-dependent, rather than universally applicable.

a: Short-Term Performance (3 Days).

GPT-4o demonstrated superior accuracy in short-term forecasting for the majority of stocks, achieving best performance in 3 out of 4 stocks: DXG (MAE: 0.50), FPT (MAE: 2.04), and VCB (MAE: 0.26). However, Google Gemini 2.0 Flash proved optimal for VCS (MAE: 0.68), indicating that even within short-term horizons, model selection should consider stock-specific characteristics.

b: Medium-Term Performance (14 Days).

Medium-term forecasting revealed a more balanced distribution of optimal models across stocks. Google Gemini 2.0 Flash excelled in FPT (MAE: 5.81) and VCS (MAE: 0.74), while GPT-4o achieved best results in DXG (MAE: 0.76), and Llama 3.1:8b performed optimally in VCB (MAE: 1.16). This heterogeneity suggests that medium-term prediction benefits from model diversity rather than a single dominant LLM.

c: Long-Term Performance (60 Days).

Contrary to initial expectations, long-term forecasting did not favor a single LLM consistently. Google Gemini 2.0 Flash achieved best performance only in DXG (MAE: 3.25), while GPT-4o excelled in VCB (MAE: 4.37), and Llama 3.1:8b proved optimal for VCS (MAE: 1.04). These findings indicate that long-term stability is not inherently tied to a specific LLM architecture, but rather depends on the interaction between model characteristics and stock-specific dynamics.

d: Practical Implications.

These results challenge the notion of a universally superior LLM for financial forecasting. Instead, they suggest that practitioners should adopt a *portfolio approach* to LLM selection, matching specific models to individual stocks and forecasting horizons based on empirical validation rather than theoretical assumptions about model capabilities.

3) C. RQ3: Superiority of Multi-Agent Systems over Traditional Baselines

Our empirical findings confirm that *Multi-Agent Systems* (MAS) significantly outperform both traditional LSTM base-

lines and Single-Agent LLM configurations, particularly within the volatility-prone Vietnamese stock market.

a: vs. LSTM Baselines.

The LSTM models exhibited significant stochastic instability and pronounced amplitude distortion. These models frequently overshot during bullish market regimes and undershot during bearish corrections. Such behavior underscores the inherent limitations of purely data-driven recurrent architectures in adapting to abrupt regime shifts, particularly those driven predominantly by retail investor sentiment.

b: vs. Single-Agent LLMs.

While Single-Agent architectures performed adequately for relatively stable assets (e.g., VCS), they lacked robust error-correction mechanisms under high volatility. This limitation manifested as *lag-induced inertia*, where predictions failed to adjust rapidly during abrupt regime transitions.

c: The MAS Advantage.

In contrast, Multi-Agent architectures—particularly the *Ensemble Voting* mechanism—acted as a *stability anchor*. By aggregating heterogeneous perspectives (Fundamental, Technical, Risk, and News agents), MAS effectively filtered idiosyncratic noise that plagued single-model approaches. This design proved especially effective in a market where approximately 85–90% of trading volume is driven by retail investor sentiment.

IX. LIMITATIONS AND FUTURE WORK

A. CURRENT LIMITATIONS

Several limitations must be acknowledged when interpreting these results. First, the evaluation scope encompasses four stocks from the Vietnamese market over a specific temporal period, which may constrain the generalizability of findings to other emerging markets or broader equity universes. While the selected stocks represent key economic sectors, the limited sample size may not capture the full spectrum of market behaviors and sectoral dynamics present in Vietnam's broader equity landscape [48].

Second, the study focuses primarily on price prediction accuracy metrics (MAE, RMSE, MAPE) without incorporating transaction costs, liquidity constraints, or market impact considerations that are crucial for practical trading implementations. Vietnam's T+2.5 settlement period and relatively constrained liquidity compared to developed markets introduce practical constraints not fully reflected in the evaluation framework. Additionally, the impact of bid-ask spreads, market depth variations, and execution timing on actual trading performance remains unexplored.

The framework's reliance on cloud-based LLM services introduces potential operational risks including latency variability, service availability concerns, and cost scalability issues that could affect real-time trading applications [49]. Network connectivity disruptions, API rate limitations, and service outages could compromise system reliability in produc-

tion environments. Furthermore, the cost structure of cloud-based inference may become prohibitive for high-frequency trading applications or resource-constrained institutional investors.

Third, the framework's performance during extreme market conditions—financial crises, high-impact events, or periods of extreme volatility—remains untested. The evaluation period, while including normal market volatility, may not encompass the full range of stress scenarios that could challenge the system's robustness. The coordination mechanisms' behavior during periods of extreme correlation breakdown, liquidity crises, or regulatory interventions requires further investigation to establish operational reliability under adverse conditions [50].

X. CONCLUSION

We investigate the problem of stock price forecasting in Vietnam's emerging and retail-dominated market through the validation of a specialized *Multi-Agent* forecasting framework. Based on a comprehensive empirical evaluation conducted on four representative stocks spanning multiple sectors (VCB, FPT, DXG, and VCS), the study provides clear and structured answers to its guiding research questions.

First, in addressing **RQ1 (Architectural Effectiveness Across Horizons)**, the results demonstrate that no single coordination architecture is universally optimal across all forecasting horizons. Instead, architectural effectiveness is strongly horizon-dependent. For the medium-term horizon (14 days), *Ensemble Voting* consistently delivers the most robust performance, achieving the lowest Mean Absolute Percentage Errors (MAPEs) across multiple stocks by effectively reconciling heterogeneous technical, fundamental, and sentiment-driven signals. In contrast, short-term forecasting (3 days), particularly for highly volatile assets, benefits from a *Hierarchical* coordination mechanism, where centralized decision-making suppresses speculative noise within the $T + 2.5$ settlement cycle. For long-term horizons (60 days), the optimal architecture diverges by sector, with *Hierarchical* coordination proving effective for highly regulated industries such as banking, while *Round Robin* refinement better captures cyclical dynamics in trend-driven sectors such as industrial construction.

Second, in response to **RQ2 (Impact of Foundation Models)**, the study establishes that the choice of the underlying Large Language Model (LLM) is highly context-dependent rather than universally applicable. While GPT-4o exhibits superior performance in short-term forecasting for most evaluated stocks, medium- and long-term horizons reveal a heterogeneous landscape in which Google Gemini 2.0 Flash, GPT-4o, and Llama 3.1:8b each emerge as optimal under different stock-specific conditions. These findings challenge the assumption of a single dominant LLM for financial forecasting and instead support a portfolio-based approach to LLM selection, where models are empirically matched to specific assets and forecasting horizons.

Third, in answering **RQ3 (Superiority of Multi-Agent**

Systems over Traditional Baselines), the empirical results confirm that *Multi-Agent Systems* significantly outperform both traditional LSTM baselines and Single-Agent LLM configurations. LSTM models exhibit pronounced instability under regime shifts, frequently overshooting during bullish phases and undershooting during bearish corrections, reflecting their limited capacity to incorporate market sentiment and structural changes. Single-Agent LLMs, while effective for relatively stable assets, lack robust error-correction mechanisms under high volatility. In contrast, Multi-Agent architectures—particularly Ensemble-based systems—act as stability anchors by aggregating diverse perspectives from fundamental, technical, risk, and news-oriented agents, thereby filtering idiosyncratic noise prevalent in sentiment-driven markets.

In summary, this work contributes a *horizon-adaptive Multi-Agent forecasting framework* to the field of financial artificial intelligence. The findings indicate that, within emerging markets such as Vietnam, predictive accuracy depends not solely on model complexity, but on the strategic alignment between coordination architecture, forecasting horizon, and the underlying cognitive engine. This insight provides practical guidance for both researchers and practitioners seeking robust and adaptive forecasting solutions in volatile, retail-driven financial environments.

REFERENCES

- [1] Vietnam News / FiinTrade, "Retail investors net buy over \$3 billion on stock market last year," January 2025, based on FiinTrade Market Review 2024. [Online]. Available: <https://vietnamnews.vn/economy/1690395/retail-investors-net-buy-over-3-billion-on-stock-market-last-year.html>
- [2] T. M. Ha, "High-frequency dynamics and information diffusion in the vietnamese stock market," *Journal of Economics and Business (JEB)*, 2025. [Online]. Available: <https://jeb.ueb.edu.vn/index.php/jeb/article/view/395>
- [3] World Federation of Exchanges, "Monthly reports and statistics," <https://www.world-exchanges.org/>, 2024, accessed: 2024-10-01.
- [4] C. M. Jones, "What do we know about high-frequency trading?" *Journal of Financial Markets*, vol. 46, pp. 100–120, 2019.
- [5] Ho Chi Minh Stock Exchange (HOSE), "Annual market report 2024," <https://www.hsx.vn/>, 2024, accessed: 2024-10-01.
- [6] Asian Development Bank, "Asian capital markets monitor 2024," <https://www.adb.org/>, 2024, accessed: 2024-10-01.
- [7] M. H. Nguyen and Q. A. Tran, "Market microstructure and order flow dynamics in emerging markets: Evidence from vietnam," *Pacific-Basin Finance Journal*, vol. 74, p. 101725, 2022.
- [8] SSI Research, "Vietnam retail investor report," <https://www.ssi.com.vn/>, 2024, accessed: 2024-10-01.
- [9] State Bank of Vietnam, "Monetary policy statements and market impact reports," <https://www.sbv.gov.vn/>, 2023, accessed: 2024-10-01.
- [10] Bloomberg L.P., "Bloomberg terminal ai integration overview," <https://www.bloomberg.com/professional/solution/bloomberg-terminal/>, 2023, accessed: 2024-10-01.
- [11] K.-j. Kim, "Financial time series forecasting using support vector machines," *Neurocomputing*, vol. 55, no. 1–2, pp. 307–319, 2003.
- [12] W. Huang, Y. Nakamori, and S.-Y. Wang, "Forecasting stock market movement direction with support vector machine," *Computers & Operations Research*, vol. 32, no. 10, pp. 2513–2522, 2005.
- [13] J. Patel, S. Shah, P. Thakkar, and K. Kotecha, "Predicting stock market index using fusion of machine learning techniques," *Expert Systems with Applications*, vol. 42, no. 4, pp. 2162–2172, 2015.
- [14] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, 2018.
- [15] K. Chen, Y. Zhou, and F. Dai, "A lstm-based method for stock returns prediction: A case study of china stock market," in *2015 IEEE International Conference on Big Data (Big Data)*. IEEE, 2015, pp. 2823–2824.

- [16] B. LeBaron, "Agent-based computational finance," in *Handbook of Computational Economics*, L. Tesfatsion and K. L. Judd, Eds. Elsevier, 2006, vol. 2, pp. 1187–1233.
- [17] S.-H. Chen, C.-L. Chang, and Y.-R. Du, "Agent-based economic models and econometrics," *The Knowledge Engineering Review*, vol. 27, no. 2, pp. 187–219, 2012.
- [18] L. Tesfatsion and K. L. Judd, *Handbook of Computational Economics: Agent-Based Computational Economics*. Elsevier, 2006, vol. 2.
- [19] S. Wang, Y. Li, and L. Ding, "Multi-agent reinforcement learning for stock trading," *Expert Systems with Applications*, vol. 177, p. 114778, 2021.
- [20] M. T. Tran and T. T. H. Nguyen, "Forecasting stock prices using machine learning techniques: Evidence from vietnam," *Journal of Asian Finance, Economics and Business*, vol. 7, no. 4, pp. 123–131, 2020.
- [21] Q. H. Pham and V. N. Le, "Stock price prediction in vietnam using hybrid lstm deep learning model," *Computational Economics*, vol. 61, no. 2, pp. 451–472, 2023.
- [22] S.-V. Oprea, A. Băra, A.-I. Andreescu, and M. P. Cristescu, "Conceptual architecture of a blockchain solution for e-voting in elections at the university level," *IEEE Access*, vol. 11, pp. 18 461–18 474, 2023.
- [23] T. McAndrew et al., "Adaptively stacking ensembles for influenza forecasting," *PLoS ONE*, vol. 16, p. e0248111, 2021.
- [24] J. Lu, Y. Fang, and W. Han, "A novel adaptive-weight ensemble surrogate model based on distance and mixture error," *PLOS ONE*, vol. 18, no. 10, p. e0293318, 2023.
- [25] J. Li, X. Zhang, and Z. Liu, "Hybrid ensemble learning for financial time-series forecasting," *Applied Intelligence*, vol. 52, no. 5, pp. 5293–5310, 2022.
- [26] Tin nhanh Chng khoán, "Phat trien ben vung thi truong tu co cau nha dau tu," *Tap chi Dau tu Chung khoan*, may 2025, accessed: 2025-10-30. [Online]. Available: <https://www.tinnhanhchungkhoan.vn/phat-trien-ben-vung-thi-truong-tu-co-cau-nha-dau-tu-post369022.html>
- [27] Tap chi Kinh te va Tai chinh, "Ty le nha dau tu ca nhan lon de tac dong tieu cuc toi thi truong chung khoan," *Tap chi Kinh te va Tai chinh Viet Nam*, aug 2024, citing World Bank (WB) August 2024 Report. Accessed: 2025-10-30. [Online]. Available: <https://tapchikinhthetachinh.vn/ty-le-nha-dau-tu-ca-nhan-lon-de-tac-dong-tieu-cuc-toi-thi-truong-chung-khoan.html>
- [28] Yuanta Viet Nam, "He luy tu ty le nha dau tu ca nhan lon," *Yuanta Viet Nam*, jan 2025, accessed: 2025-10-30. [Online]. Available: <https://yuanta.com.vn/tin-tuc/he-luy-tu-ty-le-nha-dau-tu-ca-nhan-lon>
- [29] C. J. Willmott and K. Matsuura, "Advantages of the mean absolute error (mae) over the root mean square error (rmse) in assessing average model performance," *Climate Research*, vol. 30, no. 1, pp. 79–82, 2005.
- [30] T. Chai and R. R. Draxler, "Root mean square error (rmse) or mean absolute error (mae)?—arguments against avoiding rmse in the literature," *Geoscientific Model Development*, vol. 7, no. 3, pp. 1247–1250, 2014.
- [31] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.
- [32] Z.-H. Zhou, *Ensemble Methods: Foundations and Algorithms*. Chapman and Hall/CRC, 2012.
- [33] L. Breiman, "Bagging predictors," *Machine Learning*, vol. 24, no. 2, pp. 123–140, 1996.
- [34] T. G. Dietterich, "Ensemble methods in machine learning," *International Workshop on Multiple Classifier Systems*, pp. 1–15, 2000.
- [35] X. V. Vo, "Financial market efficiency in emerging markets: Evidence from vietnam," *Finance Research Letters*, 2019.
- [36] T. Nguyen and M. Tran, "Market microstructure and retail trading in vietnam," *Asian Economic Papers*, 2021.
- [37] G. Bekaert, C. R. Harvey, C. T. Lundblad, and S. Siegel, "The european union, the euro, and equity market integration," *Journal of Financial Economics*, vol. 86, no. 3, pp. 606–644, 2007.
- [38] J. Y. Campbell, A. W. Lo, and A. C. MacKinlay, *The econometrics of financial markets*. Princeton University Press, 1997.
- [39] M. Guidolin and A. Timmermann, "Asset allocation under multivariate regime switching," *Journal of Economic Dynamics and Control*, vol. 31, no. 11, pp. 3503–3544, 2007.
- [40] J. D. M.-W. C. Kenton and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2019.
- [41] P. C. Tetlock, "Giving content to investor sentiment: The role of media in the stock market," *The Journal of Finance*, vol. 62, no. 3, pp. 1139–1168, 2007.
- [42] L. K. Hansen and P. Salamon, "Neural network ensembles," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 12, no. 10, pp. 993–1001, 1990.
- [43] C. R. Harvey, "Predictable risk and returns in emerging markets," *The Review of Financial Studies*, vol. 8, no. 3, pp. 773–816, 1995.
- [44] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–37, 2014.
- [45] R. e. a. Bommasani, "On the opportunities and risks of foundation models," *arXiv preprint arXiv:2108.07258*, 2021.
- [46] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell et al., "Language models are few-shot learners," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.
- [47] A. B. Arrieta, N. Díez-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. García, S. Gil-López, D. Molina, R. Benjamins et al., "Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai," *Information Fusion*, vol. 58, pp. 82–115, 2020.
- [48] D. K. Phan et al., "Stock market development and macro-economic factors: Evidence from vietnam," *Journal of Asian Finance, Economics and Business*, vol. 2, no. 3, pp. 27–38, 2015.
- [49] X. Qiu, T. Sun, Y. Xu, Y. Shao, N. Dai, and X. Huang, "Pre-trained models for natural language processing: A survey," *Science China Technological Sciences*, vol. 63, no. 10, pp. 1872–1897, 2020.
- [50] R. Cont, "Empirical properties of asset returns: Stylized facts and statistical issues," *Quantitative Finance*, vol. 1, no. 2, pp. 223–236, 2001.

...